

Project 2: Exploratory Data Analysis (EDA)

DecodeLabs Data Analytics Internship — Industrial Training Kit

Dataset: Dataset_for_Data_Analytics_CLEANED.xlsx | 1,200 records | 2023–2025

Tools: Microsoft Excel (PivotTables, PivotCharts, statistical functions)

1. Problem Statement

Following Project 1's data cleaning phase, this analysis investigates the underlying patterns, trends, and distributions within the transactional dataset. The core question: what does this data reveal about order value, product performance, revenue trends over time, and the relationships between key variables — and are there any outliers that represent data errors (noise) versus genuine business signals?

2. Methodology

Analysis was performed entirely in Microsoft Excel on the cleaned dataset. The approach followed the IPO framework: descriptive statistics (mean, median, five-number summary) were calculated using AVERAGE(), MEDIAN(), and QUARTILE() functions; outlier detection used the IQR method ($1.5 \times \text{IQR}$ beyond Q1/Q3) via QUARTILE() formulas cross-checked with conditional formatting; PivotTables were used for categorical breakdowns and correlation checks; and PivotTables/PivotCharts were used for time-series trend analysis.

3. Key Findings

3.1 Descriptive Statistics (Five-Number Summary)

Metric	Quantity	UnitPrice (\$)	ItemsInCart	TotalPrice (\$)
Min	1	11.39	1	11.39
Q1	2	186.06	4	410.52
Median	3	364.21	5	823.62
Q3	4	521.57	7	1,578.48
Max	5	699.93	10	3,456.40
Mean	2.95	356.41	5.49	1,053.97
Std Dev	1.41	197.18	2.28	819.86

TotalPrice shows a right-skewed distribution (mean \$1,053.97 > median \$823.62), meaning a small number of high-value orders pull the average upward. This confirms median is the more reliable central tendency measure for order value.

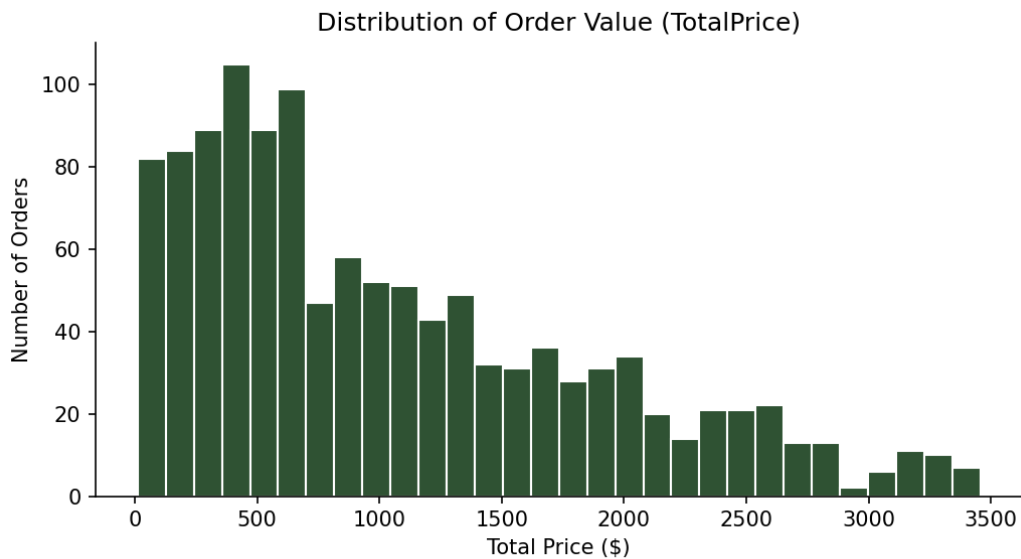


Fig 1: Distribution of Order Value — right-skewed shape confirmed.

3.2 Outlier Detection

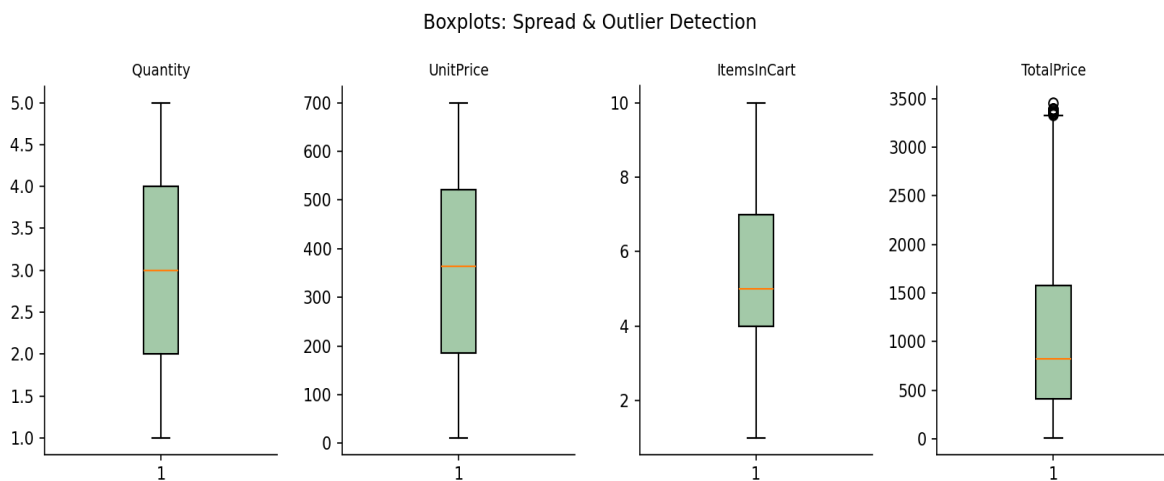


Fig 2: Boxplots across all numeric fields.

Using the IQR method, Quantity, UnitPrice, and ItemsInCart show zero outliers — all values fall within expected operational ranges. TotalPrice shows 8 outliers (values above \$3,330.41). Investigation confirms these are **signal, not noise**: each represents a legitimate maximum-quantity (5 units) order of a high-unit-price product (e.g. Printer, Laptop, Tablet near \$670–\$699 each) — not data entry errors. No cleaning action is required for these records.

3.3 Revenue by Product

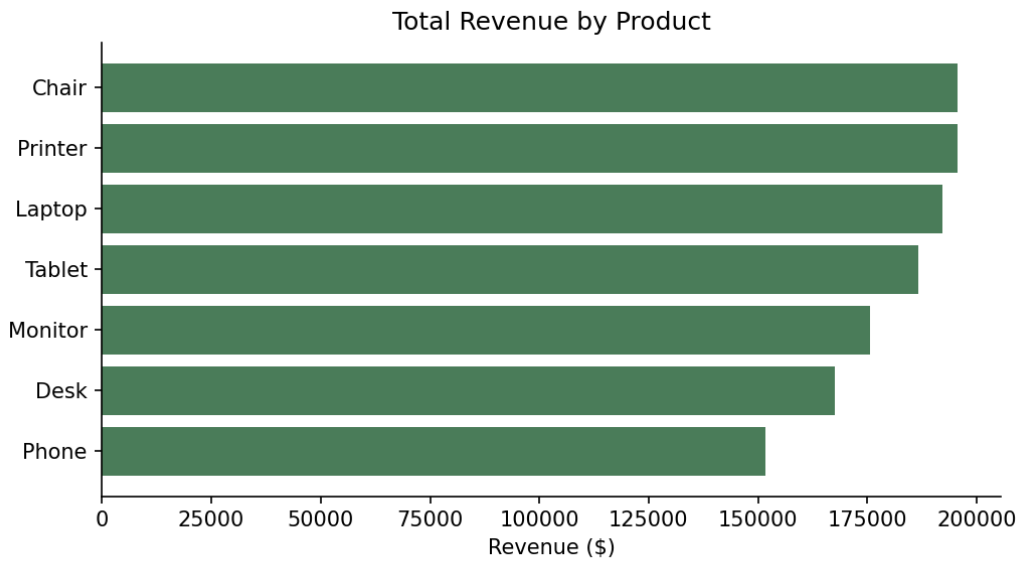


Fig 3: Total revenue contribution by product category.

Chair (\$195,620) and Printer (\$195,613) are the top revenue generators, closely followed by Laptop (\$192,127). Phone (\$151,722) contributes the least, despite being ordered a comparable number of times — driven by its lower average unit price relative to other categories.

3.4 Revenue Trend Over Time

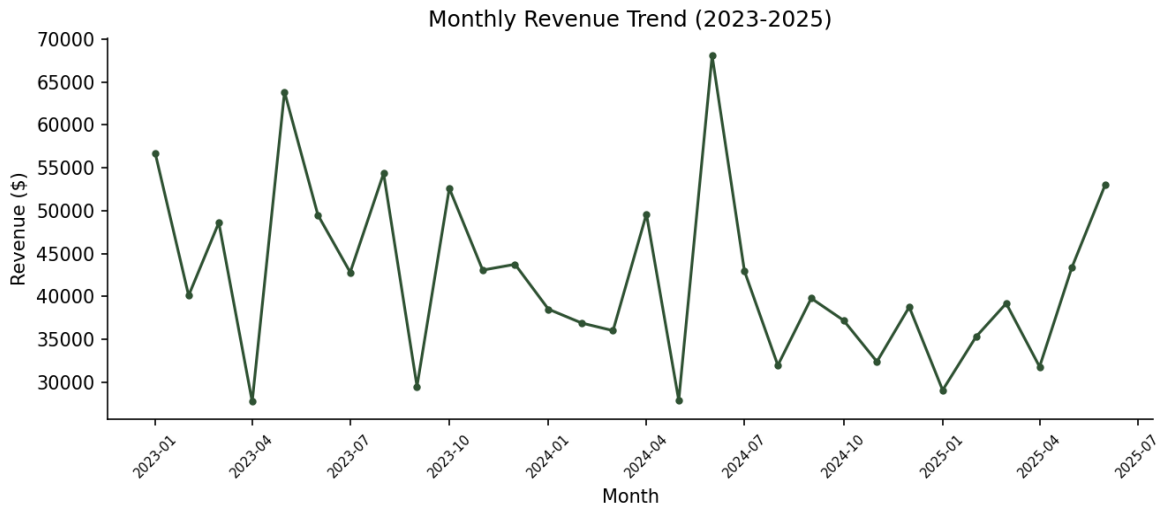


Fig 4: Monthly revenue, Jan 2023 – Jun 2025.

Revenue fluctuates month-to-month without a strong sustained upward or downward trend, suggesting order volume is relatively stable across the observed period rather than seasonal or growth-driven.

3.5 Correlation Analysis

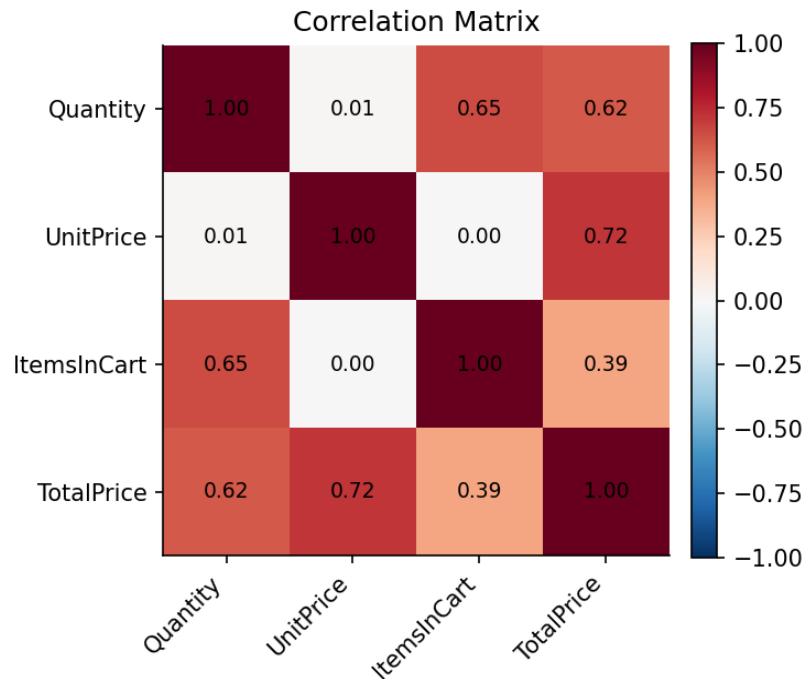


Fig 5: CORREL() matrix of numeric fields, calculated in Excel.

UnitPrice has the strongest correlation with TotalPrice ($r = 0.72$), followed by Quantity ($r = 0.62$) — both expected, since TotalPrice is derived from them. Quantity and ItemsInCart show a moderate positive correlation ($r = 0.65$). UnitPrice and Quantity are essentially uncorrelated ($r = 0.01$), indicating customers do not systematically buy more units of cheaper products or vice versa.

3.6 Order Status Distribution

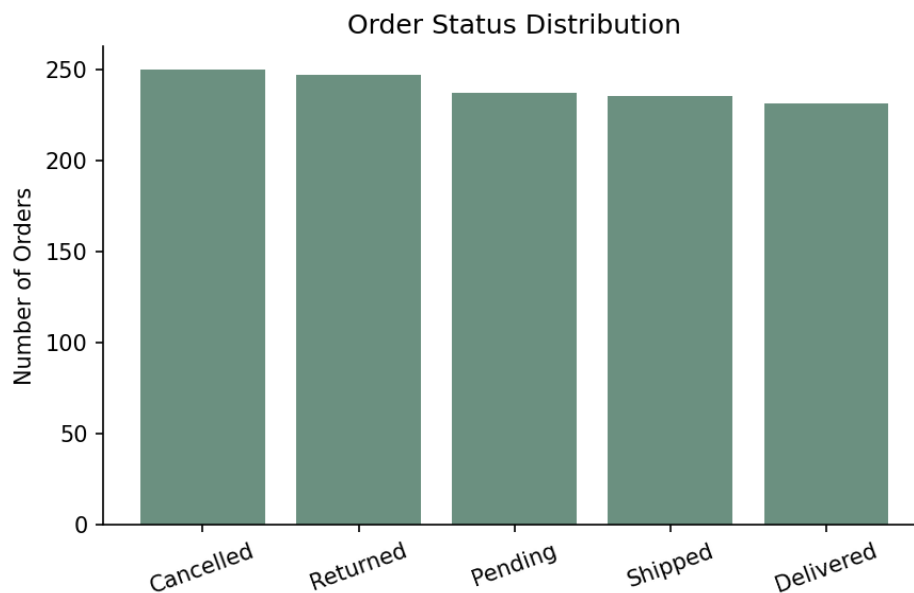


Fig 6: Order status breakdown.

Order statuses are fairly evenly distributed (231–250 orders each across Delivered, Shipped, Pending, Returned, Cancelled). Notably, Cancelled (250) and Returned (247) orders combined represent 41.4% of all orders — a

meaningful share worth flagging for further root-cause investigation.

3.7 Referral Source & Coupon Impact

Referral Source	Orders	Avg Order Value (\$)
Facebook	228	1,098.29
Instagram	259	1,062.88
Email	250	1,047.23
Google	241	1,039.18
Referral	222	1,021.69

Instagram drives the highest order volume (259), while Facebook customers spend the most per order on average (\$1,098.29). Referral-based customers have both the lowest volume and lowest average order value, suggesting this channel may need a different engagement strategy. Across coupon types, average order value is broadly similar (\$1,036–\$1,070), indicating coupons are not strongly cannibalizing order size.

4. Recommendations

- Investigate the 41.4% combined Cancelled + Returned rate — a root-cause analysis (e.g. by product or payment method) could recover meaningful revenue.
- Double down on Facebook and Instagram acquisition channels, which show the highest order volume and value respectively.
- Re-evaluate the Referral program, given its comparatively lower order volume and value.
- The 8 high-value outlier orders are legitimate bulk purchases — no data cleaning action needed, but they may represent a segment worth targeting with bulk-order promotions.
- Since UnitPrice is the strongest driver of TotalPrice, pricing strategy has more revenue leverage than encouraging higher unit quantities per order.

5. Conclusion

This EDA translated 1,200 raw transaction records into a set of verified, actionable insights: order value is right-skewed and driven primarily by unit price; the 8 statistical outliers are genuine high-value orders rather than data errors; and channel-level differences in order volume and value point to specific areas (cancellations/returns, referral engagement) for further business investigation.